1. Consider the stream function (ψ) and vorticity (ω) equations for solving a fluid flow given below.

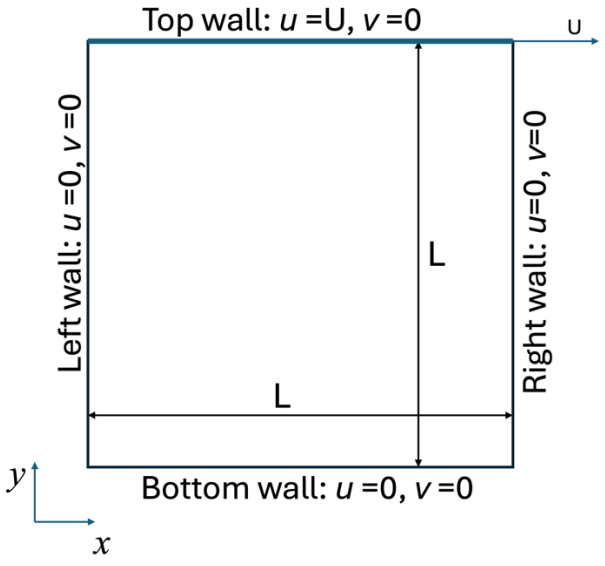
$$\frac{\partial^2\psi}{\partial x^2}+\frac{\partial^2\psi}{\partial \mathrm{y}^2}=-\omega \tag{1}$$

$$\frac{\partial\psi}{\partial \mathrm{y}}\frac{\partial\omega}{\partial \mathrm{x}}-\frac{\partial\psi}{\partial \mathrm{x}}\frac{\partial\omega}{\partial \mathrm{y}}=\upsilon\left(\frac{\partial^2\omega}{\partial x^2}+\frac{\partial^2\omega}{\partial \mathrm{y}^2}\right) \tag{2}$$

Here the velocity components (u,v) along x and y -directions can be calculated from the stream function as $\mathrm{u}=\partial\psi/\partial \mathrm{y}$ and $\mathrm{v}=-\partial\psi/\partial \mathrm{x}$, respectively. The two coupled equations can be solved together to find the values of ψ and ω, and eventually u and v. Moreover,

$$\omega=\frac{\partial \mathrm{v}}{\partial \mathrm{x}}-\frac{\partial \mathrm{u}}{\partial \mathrm{y}}$$

a) Discretize both equations using central difference scheme and obtain the explicit expression of ψ and ω from Eqs. (1) and (2), respectively. Use uniform grid $\Delta \mathrm{x}=\Delta \mathrm{y}$.
b) Consider a standard lid driven cavity problem in 2D where the top lid is driven at a velocity u = U along x-direction and the cavity is a square of side length L (see the figure below). All the walls have no-slip and no-penetration boundary condition (i.e. both u and v are zero at the walls). Write the appropriate boundary conditions for both ψ and ω at all the four walls. Discretize the boundary conditions and write the discretized equations for the boundary conditions.
c) Solve the discretized equations of (1) and (2) along with the appropriate boundary conditions to find the ψ and ω using point Gauss-Seidel method. Consider two values of Reynolds number to find the solution: Re = 10 and 100 . Here, Reynolds number is defined as $\mathrm{Re}=\mathrm{UL}/\upsilon$. Plot the vorticity field, streamlines and velocity vector for all the three values of Re and make a comparison.
d) Compare the results of (c) for two different grids: 100x100 and 400x400. Also compare the sensitivity of number of iterations with tolerance limits.



You are free to use any plotting tool for plotting the vorticity contour, streamlines, velocity vector and any other graphs. Please prepare a report on your data, your understanding and analysis, and submit the report along with the code.

(Hints: ψ may be assumed zero at the walls. You may use initial guess of $\psi = \omega = 0$. You may consider $U = L = 1$ and $\nu = 1/Re$ for the solution.)